Cloud Computing

Exercise-7

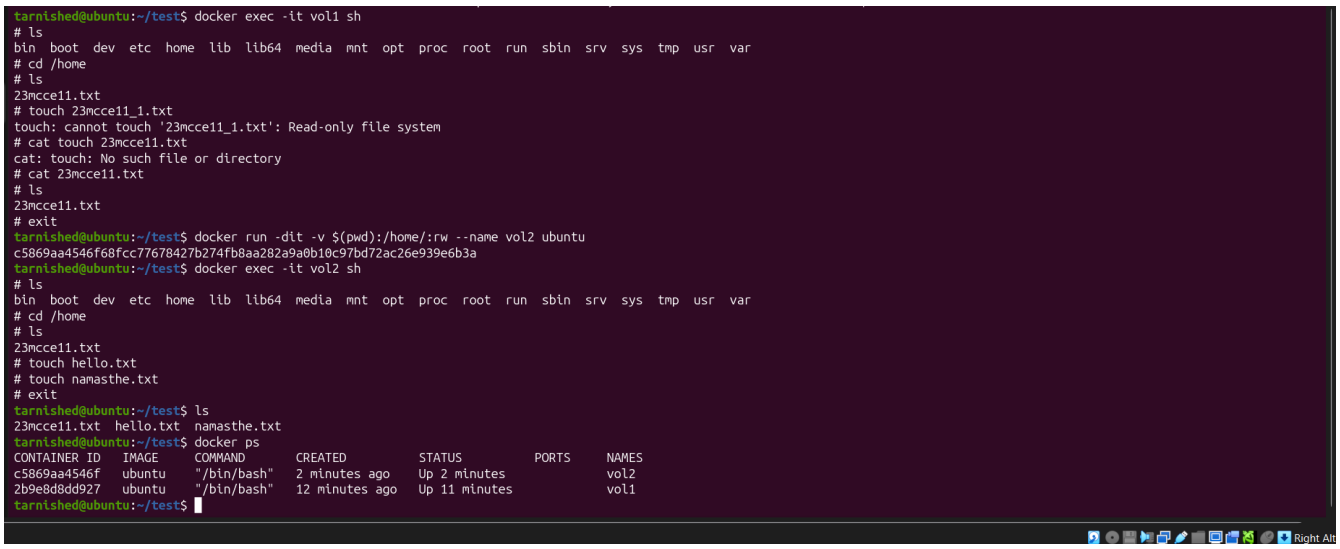
Name: CH. Aravind Kumar

ID: 23MCCE11

Course: Imtech

1.

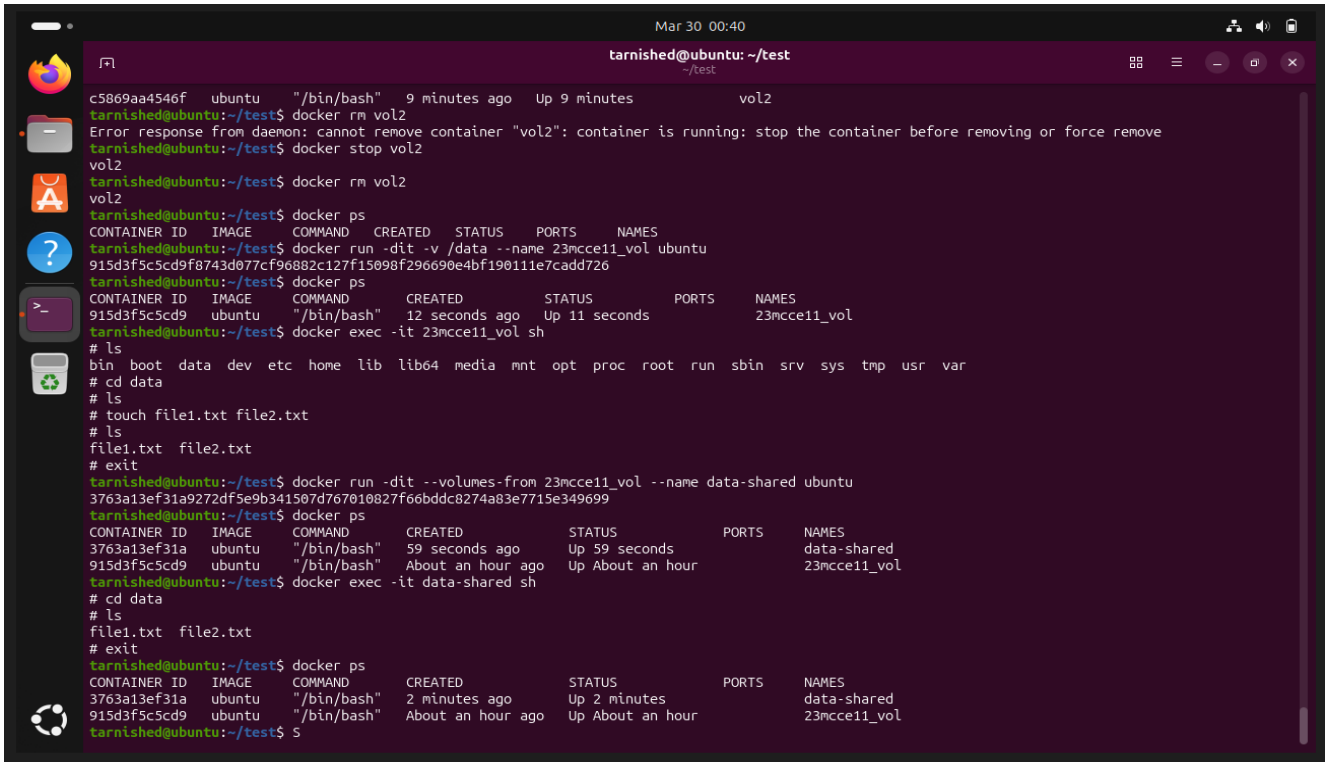
Host directory as a data volume:



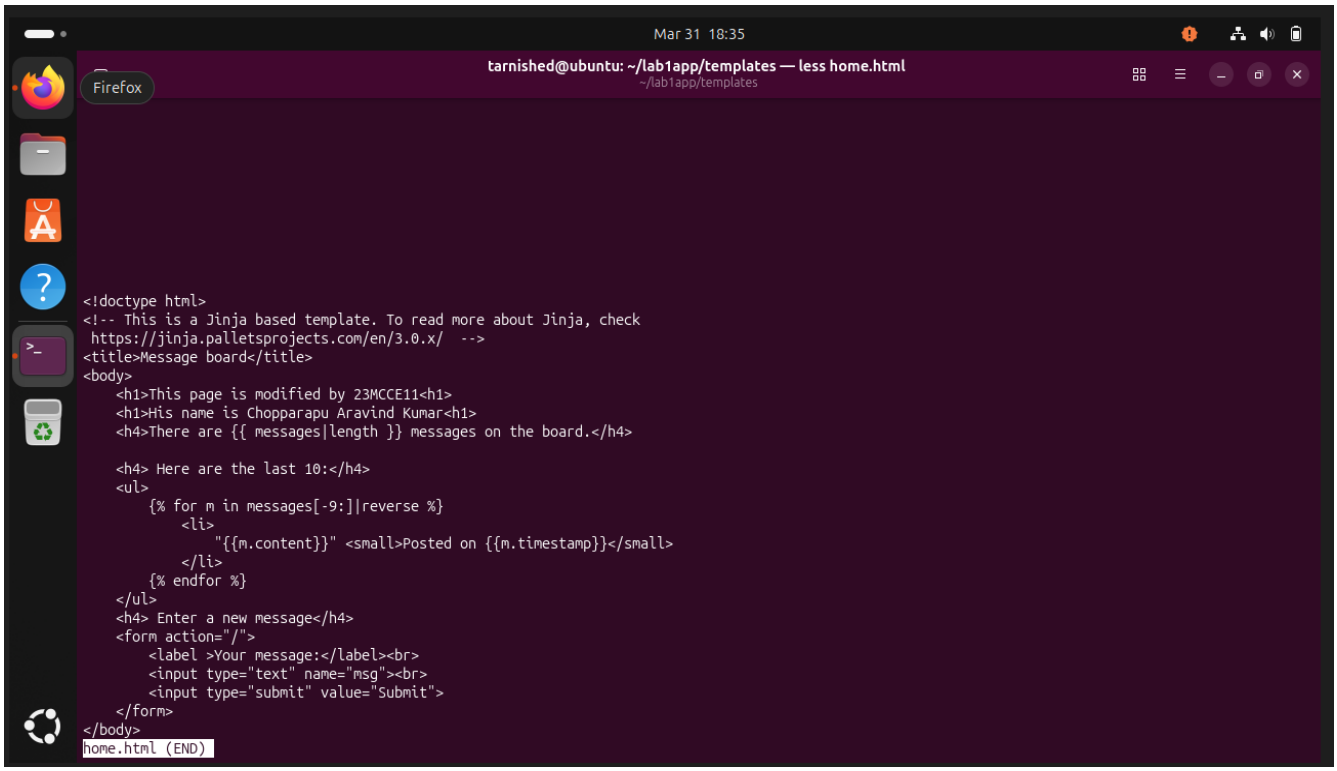
After creating the vol1, when we try creating a file touch 23mcce11_1.txt in container it shows cannot touch read only system. But for vol2 we gave rw permissions so when I created hello.txt, namasthe.txt it accepted and we can see those files in host directory.

Data volume containers:

Here I created 23mcce11_vol using ubuntu image and in that we created a directory data and 2 text files after we created data_shared mounting the data of 23mcce11_vol and we see the 2 text files.



2.



To show the file is modified.



This page is modified by 23MCCE11

His name is Chopparapu Aravind Kumar

There are 4 messages on the board.

Here are the last 10:

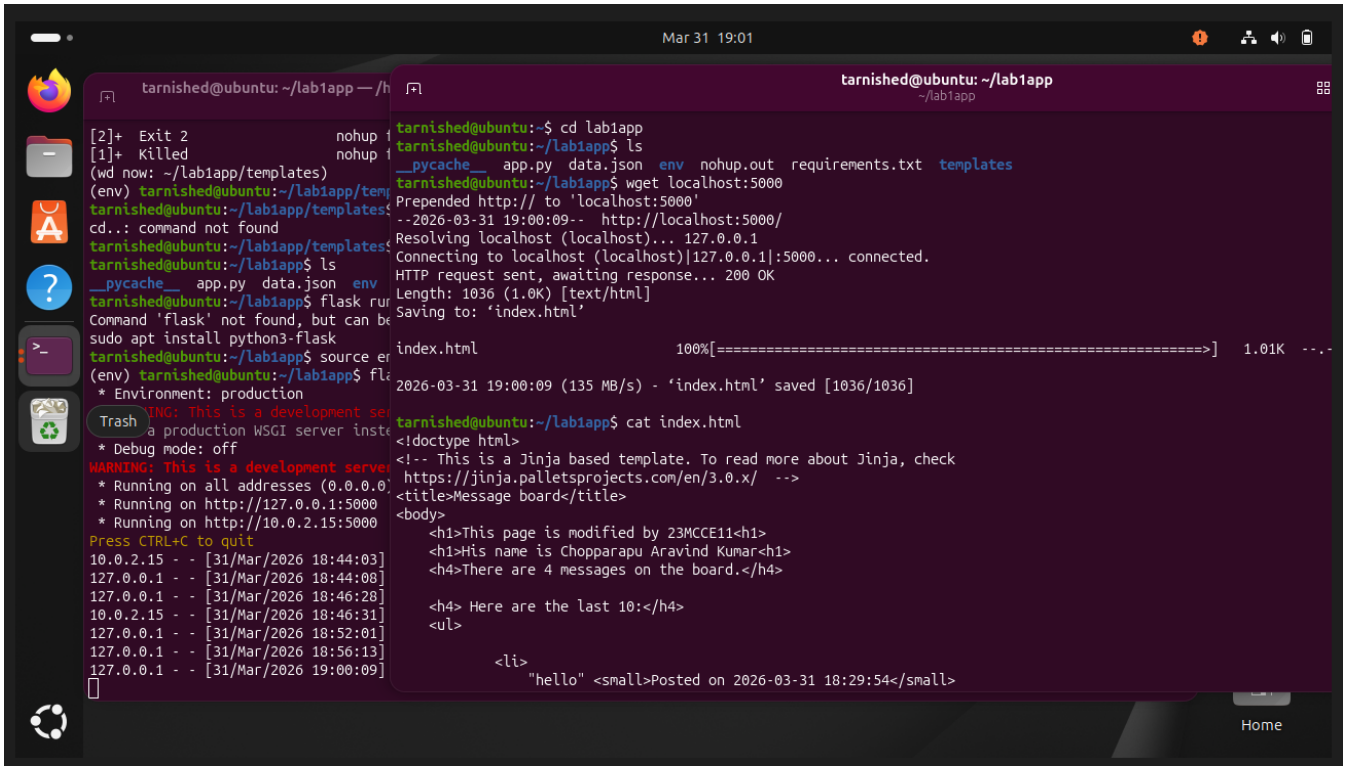
- "hello" Posted on 2026-03-31 18:29:54
- "hello" Posted on 2026-03-31 18:02:48
- "hello" Posted on 2026-03-31 18:02:34
- "hello" Posted on 2026-03-31 18:00:55

Enter a new message

Your message:

Submit

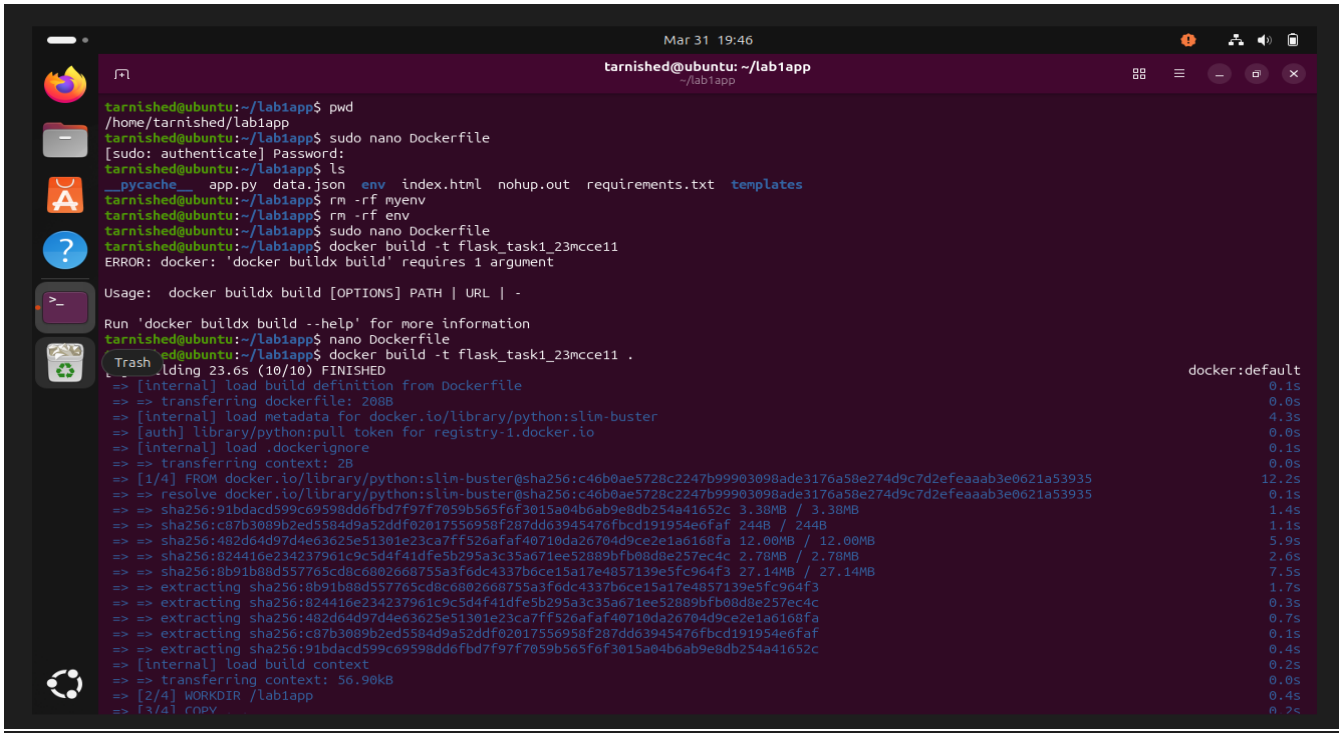
We can see the IP 10.0.2.15 which is VM IP showing the Modified page.

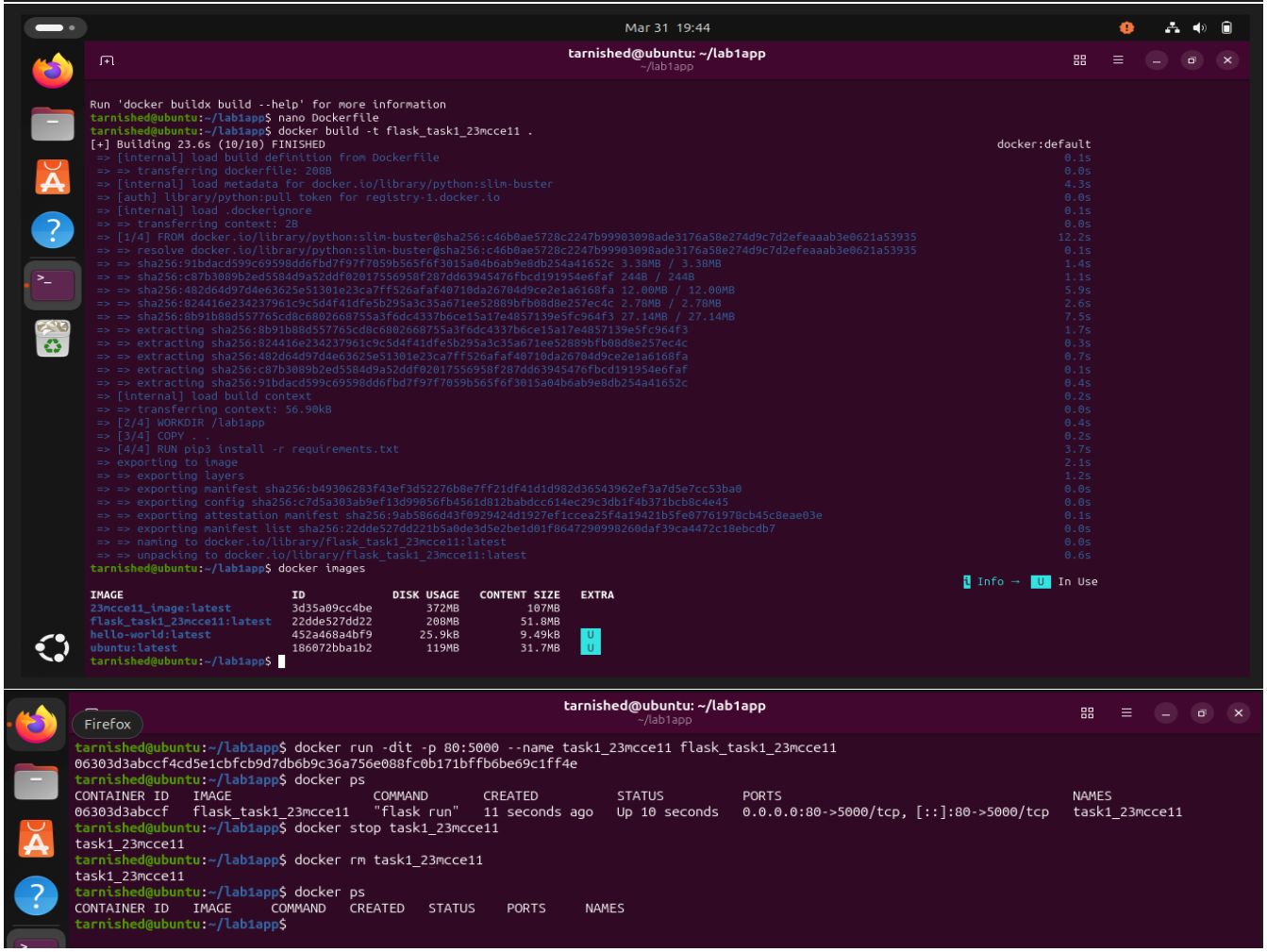


We can see 2 terminals here back one showing that server is running and front one showing the server is running with wget.

3.

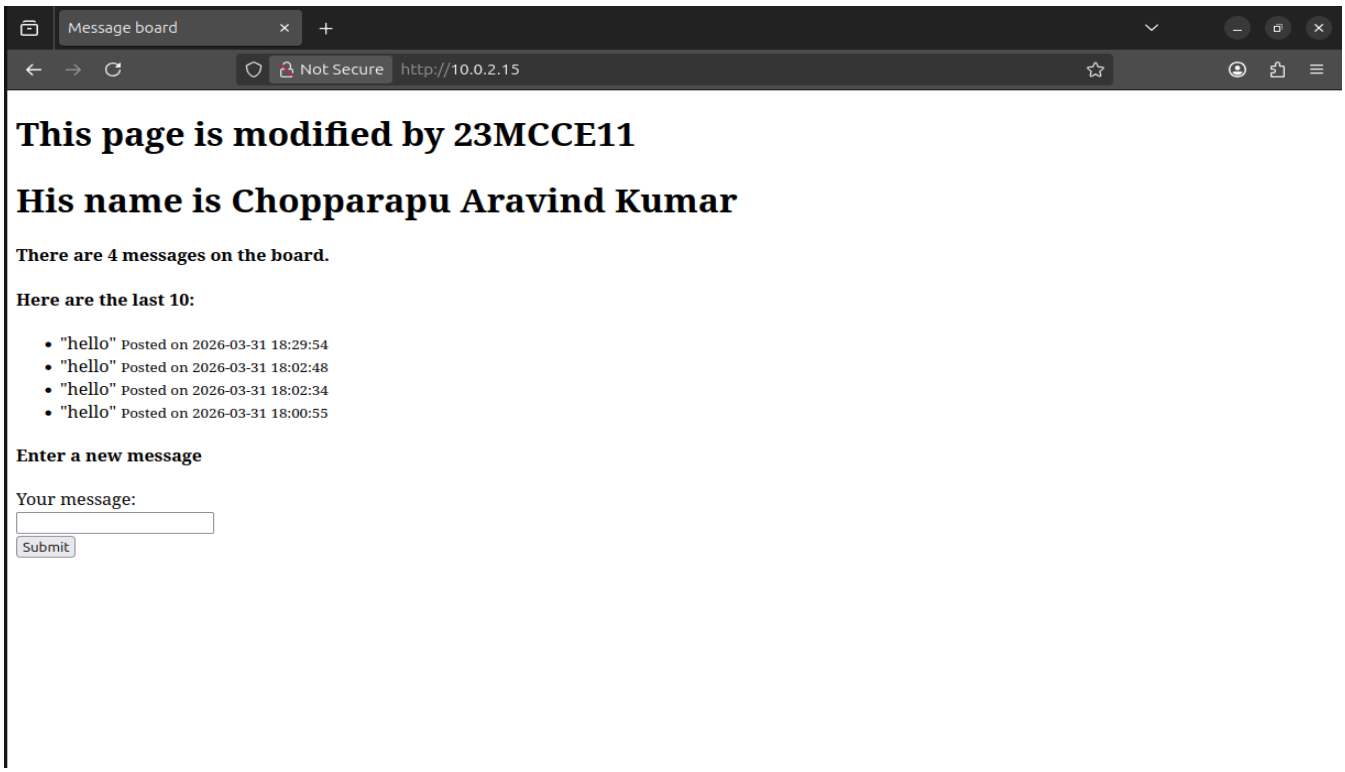
1.


Mar 31 19:46
tarnished@ubuntu: ~/lab1app
~/lab1app
tarnished@ubuntu:~/lab1app$ pwd
/home/tarnished/lab1app
tarnished@ubuntu:~/lab1app$ sudo nano Dockerfile
[sudo: authenticate] Password:
tarnished@ubuntu:~/lab1app$ ls
__pycache__ app.py data.json env index.html nohup.out requirements.txt templates
tarnished@ubuntu:~/lab1app$ rm -rf myenv
tarnished@ubuntu:~/lab1app$ rm -rf env
tarnished@ubuntu:~/lab1app$ sudo nano Dockerfile
tarnished@ubuntu:~/lab1app$ docker build -t flask_task1_23mcce11
ERROR: docker: 'docker buildx build' requires 1 argument
Usage: docker buildx build [OPTIONS] PATH | URL | -
Run 'docker buildx build --help' for more information
tarnished@ubuntu:~/lab1app$ nano Dockerfile
tarnished@ubuntu:~/lab1app$ docker build -t flask_task1_23mcce11 .
Trash
lding 23.6s (10/10) FINISHED docker:default
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 208B 0.0s
=> [internal] load metadata for docker.io/library/python:slim-buster 4.3s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [1/4] FROM docker.io/library/python:slim-buster@sha256:c46b0ae5728c2247b99903098ade3176a58e274d9c7d2efeaaab3e0621a53935 12.2s
=> => resolve docker.io/library/python:slim-buster@sha256:c46b0ae5728c2247b99903098ade3176a58e274d9c7d2efeaaab3e0621a53935 0.1s
=> => sha256:91bdacd599c69598dd6fbd7f97f7059b565f6f3015a04b6ab9e8db254a41652c 3.38MB / 3.38MB 1.4s
=> => sha256:c87b3089b2ed5584d9a52ddf02017556958f287dd63945476fbcd191954e6faf 244B / 244B 1.1s
=> => sha256:482d64d97d4e63625e51301e23ca7ff526afaf40710da26704d9ce2e1a6168fa 12.00MB / 12.00MB 5.9s
=> => sha256:824416e234237961c9c5d4f41dfe5b295a3c35a671ee52889bfb08d8e257ec4c 2.78MB / 2.78MB 2.6s
=> => sha256:8b91b88d557765cd8c6802668755a3f6dc4337b6ce15a17e4857139e5fc964f3 27.14MB / 27.14MB 7.5s
=> => extracting sha256:8b91b88d557765cd8c6802668755a3f6dc4337b6ce15a17e4857139e5fc964f3 1.7s
=> => extracting sha256:824416e234237961c9c5d4f41dfe5b295a3c35a671ee52889bfb08d8e257ec4c 0.3s
=> => extracting sha256:482d64d97d4e63625e51301e23ca7ff526afaf40710da26704d9ce2e1a6168fa 0.7s
=> => extracting sha256:c87b3089b2ed5584d9a52ddf02017556958f287dd63945476fbcd191954e6faf 0.1s
=> => extracting sha256:91bdacd599c69598dd6fbd7f97f7059b565f6f3015a04b6ab9e8db254a41652c 0.4s
=> [internal] load build context 0.2s
=> => transferring context: 56.90kB 0.0s
=> [2/4] WORKDIR /lab1app 0.4s
=> [3/4] COPY

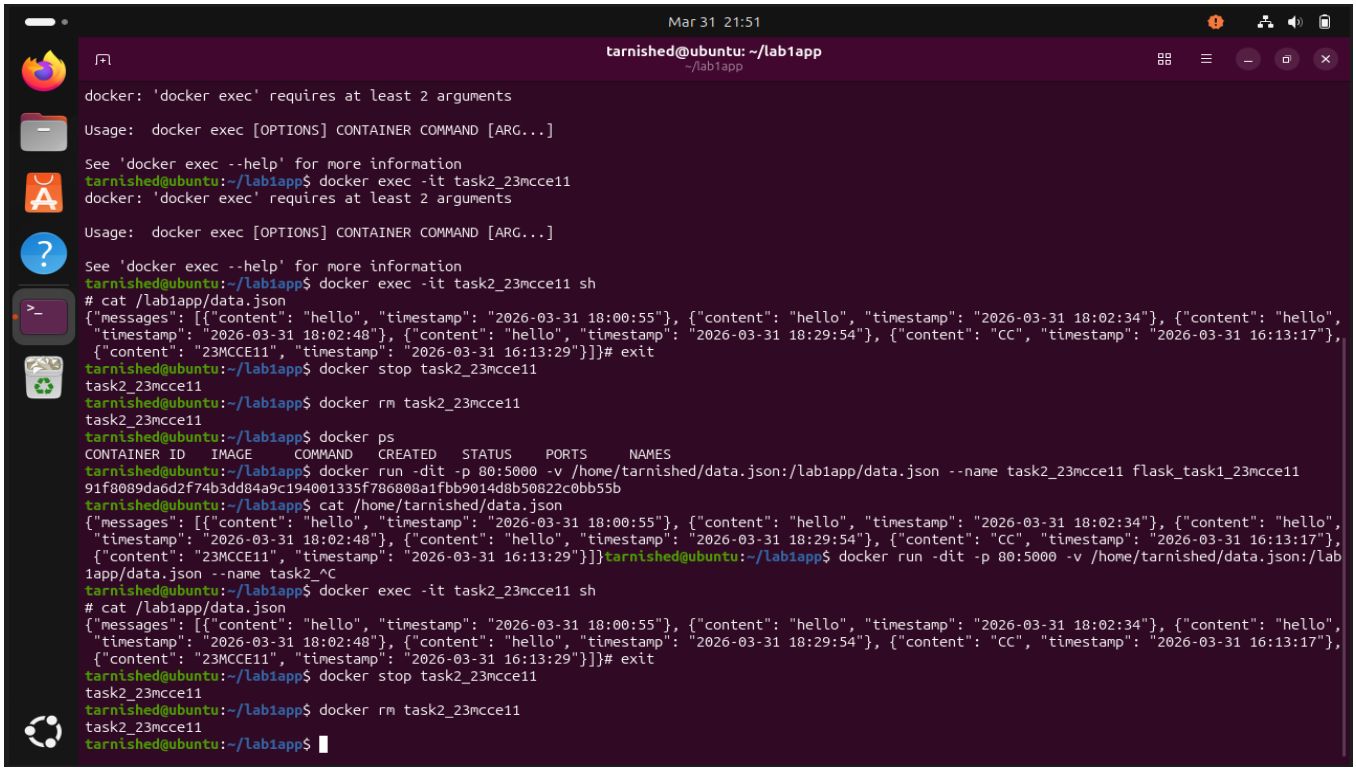

Mar 31 19:44
tarnished@ubuntu: ~/lab1app
~/lab1app
Run 'docker buildx build --help' for more information
tarnished@ubuntu:~/lab1app$ nano Dockerfile
tarnished@ubuntu:~/lab1app$ docker build -t flask_task1_23mcce11 .
[+] Building 23.6s (10/10) FINISHED docker:default
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 208B 0.0s
=> [internal] load metadata for docker.io/library/python:slim-buster 4.3s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [1/4] FROM docker.io/library/python:slim-buster@sha256:c46b0ae5728c2247b99903098ade3176a58e274d9c7d2efeaaab3e0621a53935 12.2s
=> => resolve docker.io/library/python:slim-buster@sha256:c46b0ae5728c2247b99903098ade3176a58e274d9c7d2efeaaab3e0621a53935 0.1s
=> => sha256:91bdacd599c69598dd6fbd7f97f7059b565f6f3015a04b6ab9e8db254a41652c 3.38MB / 3.38MB 1.4s
=> => sha256:c87b3089b2ed5584d9a52ddf02017556958f287dd63945476fbcd191954e6faf 244B / 244B 1.1s
=> => sha256:482d64d97d4e63625e51301e23ca7ff526afaf40710da26704d9ce2e1a6168fa 12.00MB / 12.00MB 5.9s
=> => sha256:824416e234237961c9c5d4f41dfe5b295a3c35a671ee52889bfb08d8e257ec4c 2.78MB / 2.78MB 2.6s
=> => sha256:8b91b88d557765cd8c6802668755a3f6dc4337b6ce15a17e4857139e5fc964f3 27.14MB / 27.14MB 7.5s
=> => extracting sha256:8b91b88d557765cd8c6802668755a3f6dc4337b6ce15a17e4857139e5fc964f3 1.7s
=> => extracting sha256:824416e234237961c9c5d4f41dfe5b295a3c35a671ee52889bfb08d8e257ec4c 0.3s
=> => extracting sha256:482d64d97d4e63625e51301e23ca7ff526afaf40710da26704d9ce2e1a6168fa 0.7s
=> => extracting sha256:c87b3089b2ed5584d9a52ddf02017556958f287dd63945476fbcd191954e6faf 0.1s
=> => extracting sha256:91bdacd599c69598dd6fbd7f97f7059b565f6f3015a04b6ab9e8db254a41652c 0.4s
=> [internal] load build context 0.2s
=> => transferring context: 56.90kB 0.0s
=> [2/4] WORKDIR /lab1app 0.4s
=> [3/4] COPY . . 0.2s
=> [4/4] RUN pip3 install -r requirements.txt 3.7s
=> exporting to image 2.1s
=> => exporting layers 1.2s
=> => exporting manifest sha256:b49306283f43ef3d52276b8e7ff21df41d1d982d36543962ef3a7d5e7cc53ba0 0.0s
=> => exporting config sha256:c7d5a303ab9ef13d99056fb4561d812babdcc614ec29c3db1f4b37bcb8c4e45 0.0s
=> => exporting attestation manifest sha256:9ab5866d43f0929424d1927ef1ccea25f4a19421b5fe07761978cb45c8eae03e 0.1s
=> => exporting manifest list sha256:22dde527dd221b5a8de3d5e2be1d01f8647290990260daf39ca4472c18ebcdb7 0.0s
=> => naming to docker.io/library/flask_task1_23mcce11:latest 0.0s
=> => unpacking to docker.io/library/flask_task1_23mcce11:latest 0.6s
tarnished@ubuntu:~/lab1app$ docker images
i Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
23mcce11_image:latest 3d35a09cc4be 372MB 107MB
flask_task1_23mcce11:latest 22dde527dd22 208MB 51.8MB
hello-world:latest 452a468a4bf9 25.9kB 9.49kB U
ubuntu:latest 186072bba1b2 119MB 31.7MB U
tarnished@ubuntu:~/lab1app$

Firefox
tarnished@ubuntu: ~/lab1app
~/lab1app
tarnished@ubuntu:~/lab1app$ docker run -dit -p 80:5000 --name task1_23mcce11 flask_task1_23mcce11
06303d3abccf4cd5e1cbfcb9d7db6b9c36a756e088fc0b171bffb6be69c1ff4e
tarnished@ubuntu:~/lab1app$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
06303d3abccf flask_task1_23mcce11 "flask run" 11 seconds ago Up 10 seconds 0.0.0.0:80->5000/tcp, [::]:80->5000/tcp task1_23mcce11
tarnished@ubuntu:~/lab1app$ docker stop task1_23mcce11
task1_23mcce11
tarnished@ubuntu:~/lab1app$ docker rm task1_23mcce11
task1_23mcce11
tarnished@ubuntu:~/lab1app$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
tarnished@ubuntu:~/lab1app$

Above images show the commands I used and below image show the vm ip running at 80 (even if I put http://10.0.2.15:80 it shows http://10.0.2.15).

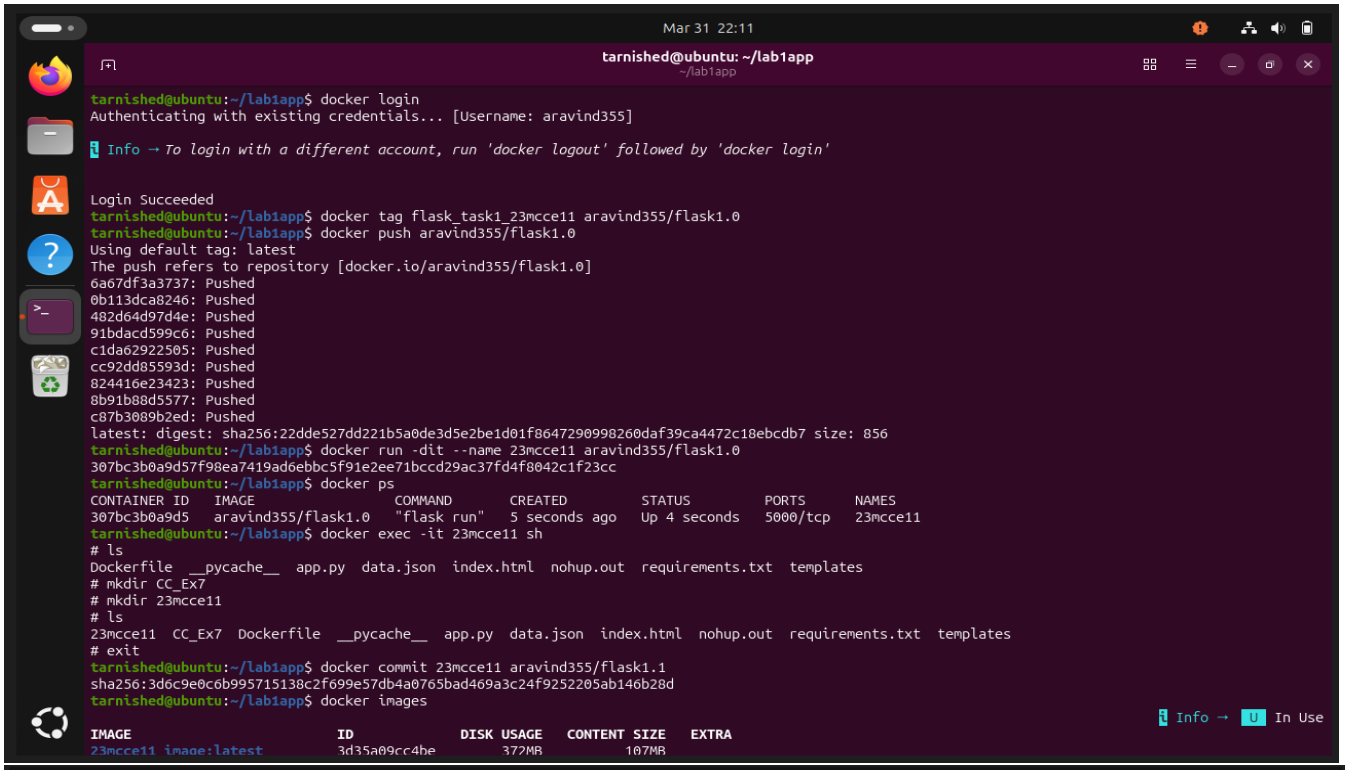


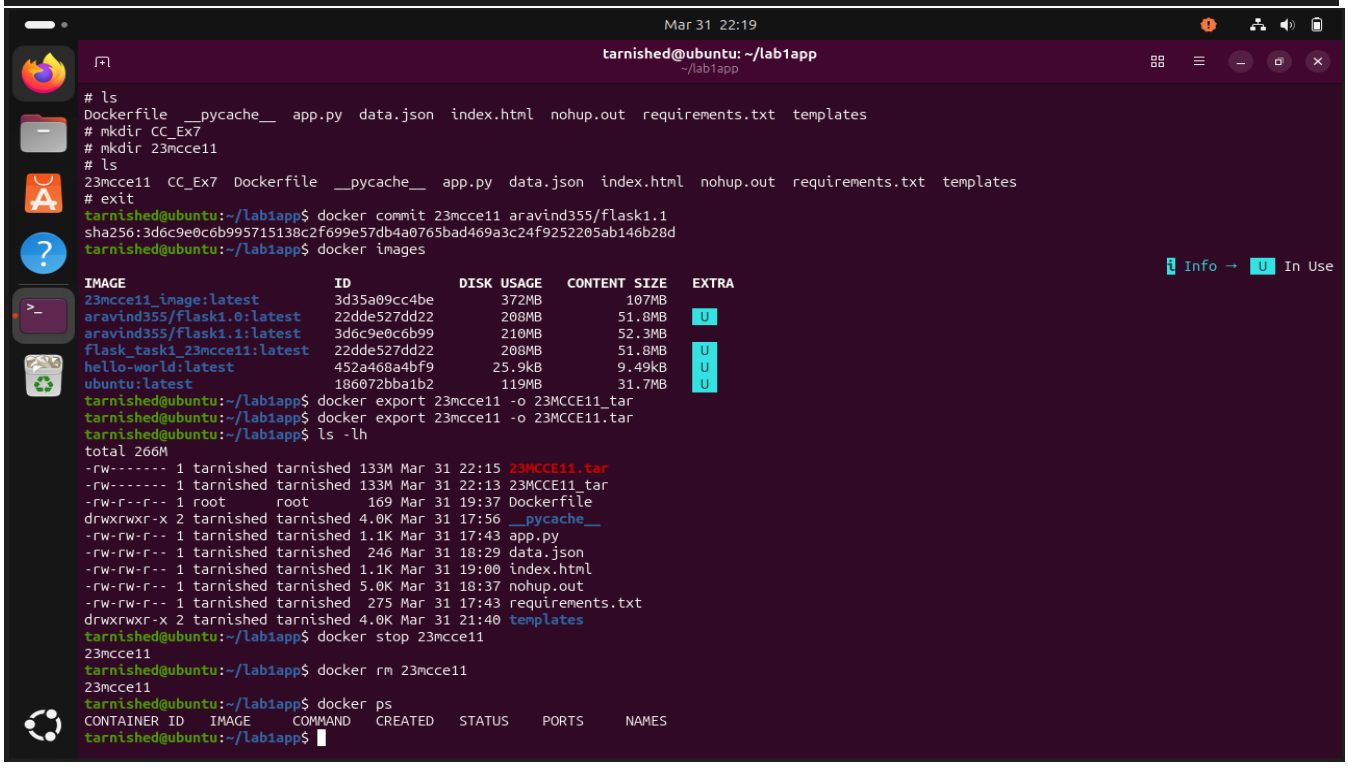
2.



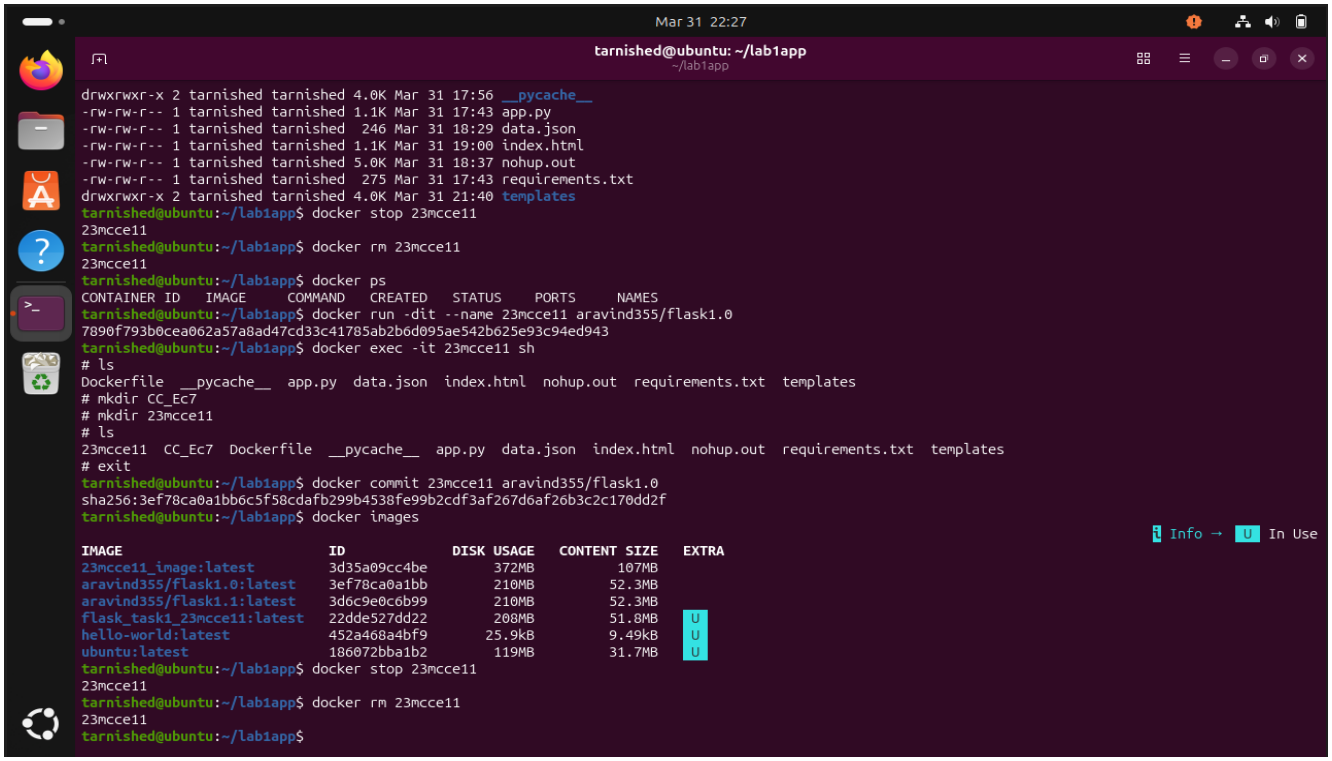
We can see the host and before and after containers data.json is same.

3.





I commit the container to a new image aravind355/flask1.1 or else we can commit it into the same image as shown below.



Above 3 show the commands and the work.

